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Toner container

Abstract

Provided a toner container having: a storage portion configured to store toner; a nozzle portion having a discharging port configured to discharge, outside the toner container, toner stored in the storage portion, the nozzle portion being fixed to the storage portion such that the nozzle portion and the storage portion are aligned in a first direction; and a loosening member configured to loose toner, the loosening member including at least a portion provided in the storage portion, wherein the loosening member is swingable relative to the storage portion about a swing axis extending in a second direction crossing the first direction.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) The present disclosure relates to a toner container used for an image forming apparatus that forms an image on a recording material.

Description of the Related Art

(2) As a method of supplying toner to an electrophotographic-type image forming apparatus, a method of supplying toner, with a toner container being mounted in a supply port that is exposed outside an image forming apparatus, has been known. CN216485993U discloses a toner container that includes a toner bottle including a stirring member inside and enables toner supply while the stirring member loosens the toner.

SUMMARY OF THE DISCLOSURE

(3) The present disclosure provides a toner container with improved performance of discharging toner.

(4) An aspect of the present disclosure provides a toner container comprising: a storage portion configured to store toner; a nozzle portion having a discharging port configured to discharge, outside the toner container, toner stored in the storage portion, the nozzle portion being fixed to the storage portion such that the nozzle portion and the storage portion are aligned in a first direction; and a loosening member configured to loose toner, the loosening member including at least a portion provided in the storage portion, wherein the loosening member is swingable relative to the storage portion about a swing axis extending in a second direction crossing the first direction.

(5) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A to 1C include sectional views of an installation portion and a toner pack according to a first embodiment during toner supply.
- (2) FIGS. 2A and 2B include a schematic sectional view of an image forming system according to the first embodiment.
- (3) FIG. 3 is a perspective view of an image forming apparatus according to the first embodiment.
- (4) FIGS. 4A and 4B include exploded perspective views of the installation portion according to the first embodiment.
- (5) FIGS. 5A and 5B include external perspective views of the installation portion according to the first embodiment.
- (6) FIGS. 6A and 6B illustrate, from above, the installation portion according to the first embodiment.
- (7) FIG. 7 illustrates, from below, the installation portion according to the first embodiment.
- (8) FIGS. 8A and 8B include perspective views of an apparatus-side shutter according to the first embodiment.
- (9) FIGS. 9A and 9B include perspective views of a cover according to the first embodiment.
- (10) FIGS. 10A and 10B include front views of the toner pack according to the first embodiment.
- (11) FIG. 11 is an exploded perspective view of the toner pack according to the first embodiment.
- (12) FIGS. 12A and 12B include a front view and a side view of a loosening member according to the first embodiment.
- (13) FIGS. 13A and 13B include perspective views of the loosening member according to the first embodiment.
- (14) FIG. 14 is a sectional view of the toner pack according to the first embodiment.
- (15) FIGS. 15A and 15B include a perspective view and a bottom view of the vicinity of a nozzle according to the first embodiment (when a pack-side shutter is closed).
- (16) FIGS. 16A and 16B include a perspective view and a bottom view of the vicinity of the nozzle according to the first embodiment (when the pack-side shutter is open).
- (17) FIG. 17 is a rear perspective view of the vicinity of the nozzle according to the first embodiment.
- (18) FIG. 18 is a front view of the vicinity of the nozzle according to the first embodiment.
- (19) FIGS. 19A and 19B include perspective views of the installation portion and the toner pack according to the first embodiment in the middle of the mounting.
- (20) FIG. 20 is a sectional view of the installation portion and the toner pack according to the first embodiment at the completion of the mounting.
- (21) FIGS. 21A and 21B include sectional views of the installation portion and the toner pack according to the first embodiment at the completion of the mounting.
- (22) FIGS. 22A to 22C include perspective views illustrating, from above, the toner pack mounted on the installation portion when an operation lever is positioned at a closing position and when at an opening position.
- (23) FIGS. 23A and 23B include sectional views of the installation portion and a toner pack according to the first embodiment.
- (24) FIGS. 24A and 24B include sectional views of the installation portion and a toner pack according to the first embodiment.
- (25) FIGS. 25A and 25B include front views of loosening members according to the first

embodiment.

(26) FIGS. **26A** and **26B** include a front view and a perspective view of a toner pack according to a second embodiment.

(27) FIGS. **27A** and **27B** include perspective views of the vicinity of a nozzle according to the second embodiment when a pack-side shutter is closed and when opened.

(28) FIGS. **28A** to **28C** include a perspective view, a front view, and a top view of a loosening member according to the second embodiment.

(29) FIG. **29** is a sectional view of the toner pack according to the second embodiment.

(30) FIGS. **30A** and **30B** include sectional views of the installation portion and the toner pack according to the second embodiment at the completion of the mounting.

(31) FIG. **31** is a sectional view of the installation portion and the toner pack according to the second embodiment.

(32) FIGS. **32A** and **32B** include a perspective view and a front view of the loosening member according to the second embodiment.

(33) FIG. **33** is a perspective view of an installation portion according to a third embodiment.

(34) FIGS. **34A** and **34B** include a top view and a sectional view of the installation portion according to the third embodiment.

(35) FIGS. **35A** and **35B** include a front view and a perspective view of a toner pack according to the third embodiment.

(36) FIGS. **36A** and **36B** include a front view and a perspective view of a loosening member according to the third embodiment.

(37) FIGS. **37A** and **37B** include a sectional view and a bottom view of the toner pack according to the third embodiment.

(38) FIG. **38** is a sectional view of the installation portion and the toner pack according to the third embodiment.

(39) FIG. **39** is an enlarged perspective view of a drive transmission portion and a driven transmission portion according to the third embodiment.

(40) FIG. **40** is a front view of a loosening member according to the third embodiment.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

(41) Hereinafter, an embodiment of the present disclosure will be described in detail based on the drawings.

(42) Image Forming System

(43) FIG. **2A** is a schematic sectional view of the configuration of an image forming system **1000** according to a first embodiment. FIG. **2B** is a perspective view of the image forming system **1000**.

(44) The image forming system **1000** includes an image forming apparatus **1**, and a toner pack **100** (toner container, toner cartridge) mountable on the image forming apparatus **1**. FIG. **3** is a perspective view of the image forming apparatus **1** on which no toner pack **100** is mounted.

(45) The toner pack **100** stores the toner that is to be supplied to the image forming apparatus **1** with the toner pack **100** being mounted on an installation portion **106** of the image forming apparatus **1** illustrated in FIG. **3**. The detailed configuration of the toner pack **100** will be described later. The toner pack **100** is moved in a mounting direction **M** illustrated in FIG. **3** to be mounted. The mounting direction **M** of the toner pack **100** is the gravitational direction in the present embodiment but may be a direction inclined relative to the gravitational direction.

(46) Image Forming Apparatus

(47) The image forming apparatus **1** is a monochrome printer that forms an image on a recording material **P** based on the image information input from an external device. Examples of the recording material **P** include papers such as a plain paper and a thick paper, a plastic film such as an overhead projector sheet, sheets having special shapes such as an envelope and an index paper, and sheet materials varying in material such as fabrics.

(48) As FIGS. 2A and 2B illustrate, the image forming apparatus **1** includes the following constituents. The constituents are an image forming device **10** that forms a toner image on the recording material **P**, a pickup roller **65** that feeds the recording material **P** to the image forming device **10**, a fixing portion **70** that fixes the toner image formed at the image forming device **10** to the recording material **P**, and a discharge roller pair **80**.

(49) The image forming device **10** includes a scanner unit **11**, a process unit **20** of electrophotographic type, and a transfer roller **12** that transfers the toner image, as a developer image formed on a photoconductor drum **21** of the process unit **20**, onto the recording material **P**. The process unit **20** includes the photoconductor drum **21**, a charging roller **22**, a pre-exposure portion **23**, and a developing device **30** (developing unit, developing portion) including a development roller **31**.

(50) The photoconductor drum **21** (image carrying member) is a photosensitive member formed into a circular cylindrical shape. The photoconductor drum **21** in the present embodiment includes a drum-shaped base body formed by using aluminum, and a photosensitive layer, on the base body, formed by a negatively charged organic photosensitive member. In addition, the photoconductor drum **21** is rotationally driven by a motor in a predetermined rotational direction (the clockwise direction in the figure) at a predetermined process speed.

(51) The charging roller **22** contacts the photoconductor drum **21** with a predetermined pressure contact force and forms a charging part. In addition, by applying a desired charging voltage by using a charging high-voltage power source, a surface of the photoconductor drum **21** is uniformly charged to a predetermined potential. In the present embodiment, the photoconductor drum **21** is charged to negative polarity by the charging roller **22**. A surface potential of the photoconductor drum **21** before reaching the charging part is eliminated by the pre-exposure portion **23** for stable discharge at the charging part.

(52) The scanner unit **11** serving as an exposure unit scans and exposes the surface of the photoconductor drum **21** by irradiating, using a polygon mirror, the photoconductor drum **21** with a laser light corresponding to the image information input from an external device. With this exposure, an electrostatic latent image corresponding to the image information is formed on the surface of the photoconductor drum **21**. Note that the scanner unit **11** is not limited to a laser scanner device, and, for example, an LED exposure device including an LED array including plural LEDs aligned in the longitudinal direction of the photoconductor drum **21** may be employed.

(53) The developing device **30** includes the development roller **31** serving as a developer carrying member that carries a developer, a developing container **32** (development frame) serving as a frame of the developing device **30**, and a supply roller **33** that can supply the development roller **31** with the developer. The development roller **31** and the supply roller **33** are supported by the developing container **32** so as to be rotatable. In addition, the development roller **31** is disposed in an opening portion of the developing container **32** so as to face the photoconductor drum **21**. The supply roller **33** is in contact with the development roller **31** so as to be rotatable, and the supply roller **33** supplies toner serving as the developer stored in the developing container **32**, onto a surface of the development roller **31**. Note that the supply roller **33** is not necessarily required as long as the toner can be supplied sufficiently to the development roller **31**.

(54) The developing device **30** in the present embodiment employs a contact developing method as a developing method. That is, a toner layer carried by the development roller **31** contacts the photoconductor drum **21** in a developing part (developing region) where the photoconductor drum **21** and the development roller **31** face each other. The development roller **31** is applied with a developing voltage by a developing high-voltage power source. Under the developing voltage, the toner carried by the development roller **31** is moved from the development roller **31** onto the drum surface according to the potential distribution of the surface of the photoconductor drum **21**, and the electrostatic latent image is developed into a toner image. Note that the present embodiment employs a reversal developing method. That is, the toner adheres to a surface region, of the

photoconductor drum **21**, whose charge amount has been attenuated by being exposed in an exposing process after charged in a charging process, thereby forming the toner image.

(55) In addition, in the present embodiment, there is used a toner, with a particle size of 6 μm , whose normal charge polarity is negative polarity. A polymerized toner produced by a polymerization method is employed as an example of the toner of the present embodiment. In addition, the toner in the present embodiment does not contain a magnetic component and is a so-called non-magnetic single component developer in which the toner is carried by the development roller **31** mainly by an intermolecular force or an electrostatic force (image force). However, a single component developer containing a magnetic component may also be used. In addition, there may be a case where a single component developer contains, in addition to toner particles, additives (examples of which include wax and silica fine particles) in order to adjust the fluidity and charging performance of the toner. In addition, a two-component developer constituted by a non-magnetic toner and a magnetic carrier may also be used as the developer. When a magnetic developer is used, for example, a developing sleeve having a circular cylindrical shape and provided with a magnet on the inner side thereof is used as the developer carrying member.

(56) The developing container **32** includes a toner storage chamber **36** (second storage portion, storage portion of main body) storing the toner. A stirring member **34** (toner conveyance member) is provided inside the toner storage chamber **36**. The stirring member **34** stirs the toner in the developing container **32** by rotating as driven by a motor, which is not illustrated, and conveys the toner toward the development roller **31** and the supply roller **33**. The stirring member **34** also functions to make the toner in the developing container uniform by circulating, in the developing container, the toner that has not been used for the development and has been scraped out of the development roller **31**. Note that the stirring member **34** is not limited to the rotating form. For example, a stirring member in a swinging form may also be employed.

(57) In addition, at the opening portion of the developing container **32** where the development roller **31** is disposed, a developing blade **35** regulating the amount of the toner carried by the development roller **31** is disposed. The toner supplied onto the surface of the development roller **31** is uniformly thinned in a layer by passing through, with the rotation of the development roller **31**, a facing part where the development roller **31** and the developing blade **35** face each other and is again charged to negative polarity by triboelectric charging.

(58) Next, an image forming operation of the image forming apparatus **1** will be described. When a command for forming an image is input to the image forming apparatus **1**, an image forming process performed by the image forming device **10** is started based on the image information input from an external computer connected to the image forming apparatus **1**. The scanner unit **11** emits a laser light toward the photoconductor drum **21** based on the input image information. At this time, the photoconductor drum **21** has been charged by the charging roller **22** in advance, and an electrostatic latent image is formed on the photoconductor drum **21** by irradiating the photoconductor drum **21** with a laser light. The electrostatic latent image is then developed by the development roller **31** into a toner image on the photoconductor drum **21**.

(59) Alongside the image forming process described above, a recording material P is fed by the pickup roller **65** and is conveyed toward a transfer nip formed by the transfer roller **12** and the photoconductor drum **21**.

(60) The transfer roller **12** is applied with a transfer voltage from a transfer high-voltage power source, and the toner image carried by the photoconductor drum **21** is transferred onto the recording material P. When the recording material P, onto which the toner image has been transferred, passes through the fixing portion **70**, the toner image is heated and applied with pressure. Thus, the toner particles are melted and subsequently adhered, and the toner image is thereby fixed on the recording material P. The recording material P that has passed through the fixing portion **70** is discharged to an outside region of the image forming apparatus **1** (outside the apparatus) by the discharge roller pair **80** serving as a discharge unit and is stacked on a discharge

tray **81** serving as a stack portion formed in an upper portion of the image forming apparatus **1**.

(61) A top cover **82** serving as a loading tray is provided in the upper portion of the image forming apparatus **1**, and the discharge tray **81** serving as a stack surface is formed on an upper surface of the top cover **82**. As FIGS. 2B and 3 illustrate, the top cover **82** is provided with an opening and closing member **83** supported so as to be openable and closable about a rotating shaft **83a** extending in a front-back direction. An opening portion **82a** that is open upward is formed in the discharge tray **81** of the top cover **82**. As FIG. 3 illustrates, the installation portion **106** provided for the mounting of the toner pack **100** is exposed through the opening portion **82a**.

(62) The opening and closing member **83** is movable between a closing position where the opening and closing member **83** covers the installation portion **106** so as to prevent the toner pack **100** from being mounted on the image forming apparatus **1** and an opening position where the opening and closing member **83** causes the installation portion **106** to be exposed so as to allow the toner pack **100** to be mounted on the image forming apparatus **1**. The opening and closing member **83** functions as a part of the discharge tray **81** when at the closing position. The opening and closing member **83** and the opening portion **82a** are formed on the left side of the discharge tray **81** when the image forming apparatus **1** is viewed from the front side. The front side of the image forming apparatus **1** described here is the upstream side of the image forming apparatus **1** in a direction where the recording material P is fed by the pickup roller **65**. In addition, a user hooks a finger into a groove portion **82b** provided in the top cover **82** and opens the opening and closing member **83** leftward.

(63) The opening portion **82a** of the discharge tray **81** is open such that the installation portion **106** formed in the upper portion of the image forming apparatus **1** is exposed, and the opening and closing member **83** is opened to allow a user to access the installation portion **106**. Note that the present embodiment employs a direct supply method where a user supplies the toner from the toner pack **100** mounted on the installation portion **106** to the developing device **30** with the developing device **30** being mounted in the image forming apparatus **1**. At least a portion of the toner pack **100** is exposed outside the image forming apparatus **1**, with the toner pack **100** being mounted on the installation portion **106** of the image forming apparatus **1**.

(64) Usability can be improved because it is not required to remove the process unit **20** from the image forming apparatus **1** and replace a new process unit when the remaining amount of the toner in the process unit **20** becomes small. In addition, the toner can be supplied into the developing container **32** at lower cost than when the entire process unit **20** is replaced. Note that, even compared with when only the developing device **30** of the process unit **20** is replaced, the direct supply method enables cost reduction because the various rollers, gears, and other parts are not required to be replaced.

(65) Installation Portion for Toner Pack

(66) First, the configuration of the installation portion **106** will be described using FIGS. 4 to 9. In the present embodiment, the installation portion **106** is a unit provided for the mounting of the toner pack **100**.

(67) FIG. 4A is an exploded perspective view of the installation portion **106**. FIG. 4B is an exploded perspective view of the installation portion **106** viewed in a direction different from that of FIG. 4A. FIGS. 5A and 6A are each a perspective view of the appearance of the installation portion **106**, viewed in a mounting direction M, when an operation lever **108** is at a closing position. FIGS. 5B and 6A are each a perspective view of the appearance of the installation portion **106**, viewed in the mounting direction M, when the operation lever **108** is at an opening position. FIG. 7 is a perspective view of the installation portion **106** viewed from the downstream side in the mounting direction M.

(68) FIG. 8A is a perspective view of an apparatus-side shutter **109** viewed from the upstream side in the mounting direction M. FIG. 8B is a perspective view of the apparatus-side shutter **109** from a point of view different from that of FIG. 8A. FIG. 9A is a perspective view of a cover **110** viewed

from the downstream side relative to the cover **110** in the mounting direction M.

(69) FIG. **9B** is a perspective view of the cover **110** viewed from the upstream side in the mounting direction M.

(70) The installation portion **106** illustrated in FIGS. **3** and **5** includes a base frame **2** including a first frame **107**, a second frame **117**, and the cover **110**. The cover **110** and the second frame **117** are fixed to the first frame **107**. As FIG. **9** illustrates, the cover **110** includes an engaged portion **110h**, and the engaged portion **110h** is engaged with an engaging portion **107b** (FIG. **4A**) of the first frame **107** such that the cover **110** is prevented from rotating about a rotational axis B relative to the first frame **107**. Note that the first frame **107**, the cover **110**, and the second frame **117** are not necessarily different members and may be configured as one body. As FIGS. **4** and **7** illustrate, the second frame **117** has an apparatus-side opening **117a** (frame opening, receiving opening), and the apparatus-side opening **117a** communicates with the toner storage chamber **36** of the developing device **30** (refer to FIG. **2A**).

(71) The operation lever **108** and the apparatus-side shutter **109** (second shutter) are each attached to the base frame **2** so as to be rotatable about the rotational axis B (center axis) relative to the base frame **2**.

(72) The first frame **107** includes a positioning portion **107a**. In a radial direction r of an imaginary circle VC centered around the rotational axis B, the positioning portion **107a** protrudes inward relative to an inner peripheral surface, of the first frame **107**, centered around the rotational axis B. In addition, the operation lever **108** includes a drive transmission portion **108a** (protruded portion of lever) and an operation portion **108b**. As FIG. **4A** illustrates, the drive transmission portion **108a** of the operation lever **108** is a protruded portion protruded inward, in the radial direction r of the imaginary circle VC centered around the rotational axis B, relative to an inner peripheral surface of the operation lever **108** centered around the rotational axis B.

(73) The apparatus-side shutter **109** is a cylindrical member whose upper side is open. As FIG. **8** illustrates, the apparatus-side shutter **109** has a receiving inlet **109a** (second shutter opening, apparatus-side shutter opening) in a side surface portion, of the apparatus-side shutter, extending in the direction of the rotational axis B, and the apparatus-side shutter **109** has a bottom surface **109b** provided with a regulated rib **109c** (rotating regulated portion). The apparatus-side shutter **109** further includes a center boss **109d** (positioning shaft, shaft portion), a driven transmission portion **109e** (pushed portion, protruded portion of apparatus-side shutter), a pack contact surface **109g** (mounting-direction positioning), and an inner peripheral surface **109h** (radial-direction positioning). The apparatus-side shutter **109** is rotatable about the rotational axis B relative to the base frame **2**.

(74) The regulated rib **109c** protrudes upward from the bottom surface **109b** in the direction of the rotational axis B. As FIG. **8A** illustrates, the driven transmission portion **109e** is a protruded portion protruded inward in the radial direction r of the imaginary circle VC centered around the rotational axis B. An apparatus-side seal **111** is stuck around the receiving inlet **109a** (refer to FIG. **5B**).

(75) Here, the apparatus-side shutter **109** is rotatable, relative to the base frame **2**, so as to take a closing position and an opening position. The receiving inlet **109a** is covered with the apparatus-side seal **111** and the cover **110** when the apparatus-side shutter **109** is at the closing position, and the receiving inlet **109a** is open without being covered with the cover **110** when the apparatus-side shutter **109** is at the opening position. The closing position is the position given in FIGS. **5A** and **6A**, and the receiving inlet **109a** of the apparatus-side shutter **109** does not communicate with the apparatus-side opening **117a** of the second frame **117** when the apparatus-side shutter **109** is at the closing position (non-communication position). The opening position is the position illustrated in FIGS. **5B** and **6B**, and the receiving inlet **109a** of the apparatus-side shutter **109** communicates with the apparatus-side opening **117a** of the second frame **117** when the apparatus-side shutter **109** is at the opening position (communication position). The apparatus-side shutter **109** moves to the opening position, and the toner can thereby be supplied (fed) from the toner pack **100** into the toner

storage chamber **36** of the developing device **30** through the receiving inlet **109a**.

(76) Note that, because the operation lever **108** and the apparatus-side shutter **109** are not coupled regarding driving, the apparatus-side shutter **109** is not rotated when the operation lever **108** is operated with the toner pack **100** not being attached.

(77) Toner Pack

(78) The basic configuration of the toner pack **100** will be described using FIGS. **10** to **14**. FIG. **10A** is a front view of the toner pack **100** when a pack-side shutter **103** is at a closing position. FIG. **10B** is a front view of the toner pack **100** when the pack-side shutter **103** is at an opening position. FIG. **11** is an exploded perspective view of the toner pack **100**. FIG. **12A** is a front view of a loosening member **200**. FIG. **12B** is a side view of the loosening member **200**. FIG. **13A** is a perspective view of the loosening member **200**. FIG. **13B** is a perspective view of the loosening member **200** in a state of being elastically deformed. FIG. **14** is a sectional view of the toner pack **100**.

(79) Toner pack **100** includes a storage portion **101** (first storage portion) storing the toner, a nozzle **102** (nozzle portion, pipe, tube, valve, discharging portion), the pack-side shutter **103** (container shutter, rotary member), and the loosening member **200** (refer to FIG. **12**). As FIG. **10** illustrates, in a first direction **D1**, the storage portion **101** is provided on one end side, and the nozzle **102** and the pack-side shutter **103** are provided on the other end side. The storage portion **101** and the nozzle **102** are aligned in the first direction **D1**.

(80) The storage portion **101** is a pouch formed by a polypropylene sheet (flexible sheet) being subjected to pouch processing. As FIGS. **10** and **11** illustrate, the storage portion **101** has a flat shape in which a width in a second direction **D2** is greater than a width in a third direction crossing both the first direction **D1** and the second direction **D2**. As FIG. **10A** illustrates, the shape includes a region of the storage portion **101** in which a width in the second direction **D2** increases as a distance from the nozzle **102** increases.

(81) The nozzle **102** in the present embodiment is formed by polypropylene. A side surface **102c** (first outer surface), of the nozzle **102**, extending in the first direction **D1** has a discharging port **102a** (nozzle opening, first opening) communicating with the inside of the storage portion **101**. The toner stored in the storage portion **101** is discharged outside the toner pack **100** through the discharging port **102a**. As FIG. **14** illustrates, the nozzle **102** has a receiving inlet **102f** that receives the toner from the storage portion **101** and a passage **102h** allowing the toner to pass through from the receiving inlet **102f** to the discharging port **102a**. In the present embodiment, the discharging port **102a** is provided in the side surface **102c** extending in the first direction **D1**.

(82) Note that the nozzle **102** and the storage portion **101** may be configured as one body. In addition, a seal may be provided between the storage portion **101** and the discharging port **102a** of the nozzle **102**, and the storage portion **101** and the discharging port **102a** may communicate with each other when the seal is removed.

(83) The pack-side shutter **103** (rotary member) is provided on the outer side relative to the side surface **102c** of the nozzle **102**. The pack-side shutter **103** is attached so as to be rotatable about a rotational axis **A** (first rotational axis) extending in the first direction **D1** and has an opening **103a** (opening of rotary member, first shutter opening) as FIG. **11** illustrates. The pack-side shutter **103** is provided on the outer side relative to the side surface **102c** in the radial direction **r** of the imaginary circle **VC** centered around the rotational axis **A**. The side surface **102c** of the nozzle **102** is a curved surface curved outward in the radial direction **r** of the imaginary circle **VC** centered around the rotational axis **A**. A surface on the inner side of the pack-side shutter **103** (surface facing the side surface **102c**) is a curved surface curved along the side surface **102c** of the nozzle **102**, and a pack-side seal **105** having a substantially rectangular shape is attached to the curved surface. Note that the side surface **102c** of the nozzle **102** is also a surface extending along the rotational axis **A**.

(84) The loosening member **200** includes, as FIG. **12A** illustrates, a base portion **201** and a first loosening portion **202** (elastic portion) formed by a wire spring (wire material).

(85) The base portion **201** is provided from the storage portion **101** to the passage **102h** of the nozzle **102** such that the longitudinal direction thereof coincides with the first direction **D1**, and the base portion **201** includes a first end portion **201a** that is one end portion in the longitudinal direction and a second end portion **201b** that is the other end portion in the longitudinal direction. The first end portion **201a** is a free end including a portion extending in the second direction **D2** orthogonal to the first direction **D1**. A second loosening portion **201b2** is provided in the second end portion **201b** so as to be positioned in the passage **102h** of the nozzle **102**.

(86) One end portion **202a** (third end portion) and the other end portion **202b** (fourth end portion) of the first loosening portion **202** are supported by the first end portion **201a** and the second end portion **201b** of the base portion **201**, respectively. As FIG. **12B** illustrates, the first loosening portion **202** has a substantially helical shape wound around the base portion **201** from the one end portion **202a** toward the other end portion **202b** so as to form an ellipse forming an angle of $\theta 1$ with a direction orthogonal to the longitudinal direction of the base portion **201**. In other words, the first loosening portion **202** is a wire spring (wire material) extending in the first direction **D1** while turning around the base portion **201**. A minor axis **W2** of the first loosening portion **202** is uniform in the longitudinal direction, whereas a major axis **W1** of the first loosening portion **202** decreases from the one end portion **202a** toward the other end portion **202b** as FIG. **12A** illustrates. In other words, in the region **al** (FIG. **10A**) of the storage portion **101** described above, a width **W3** (the major axis **W1**) of the first loosening portion **202** in the second direction **D2** increases, according to the width of the storage portion **101**, as a distance from the nozzle **102** increases. In addition, the second loosening portion **201b2** has plural protruding portions **201b21** extending in the second direction **D2** and aligned in the first direction **D1**.

(87) Because the first loosening portion **202** is formed by a wire spring, the first loosening portion **202** transitions from the state of FIG. **13A** to the state of the FIG. **13B**. The first loosening portion **202** elastically deforms so as to be displaced relative to the base portion **201** when applied with force.

(88) The second end portion **201b** of the base portion **201** is provided with a supported portion **201b1** (swing shaft portion) having a shaft shape, and supporting hole portions **102g** (supporting portions) provided in the nozzle **102** support the supported portion **201b1** as FIG. **11** illustrates. Thus, the base portion **201** is supported by the nozzle **102** so as to be swingable relative to the nozzle **102** about a swing axis **S1** of the supported portion **201b1**. When the base portion **201** swings about the swing axis **S1**, a second loosening portion **202b2** inside the passage **102h** of the nozzle **102** also swings about the swing axis **S1**. The first loosening portion **202** and the second loosening portion **202b2** are on the sides opposite from each other in the longitudinal direction of the base portion **201** relative to the supported portion **201b1**. The swing axis **S1** in the present embodiment extends in the second direction **D2** according to the shape (flat shape) of the storage portion **101**, and there is not caused the swing about an imaginary line extending in the third direction crossing both the first direction **D1** and the second direction **D2**. However, in addition to the swing about the swing axis **S1** extending in the second direction **D2**, swing may also be caused about a swing axis extending in the third direction crossing both the first direction **D1** and the second direction **D2**. Swing may be caused in every direction crossing the first direction **D1**. Note that the movement of the loosening member **200** accompanying a supply operation using the toner pack **100** and performed by a user will be described later.

(89) As FIG. **10** illustrates, the pack-side shutter **103** is rotatable about the rotational axis **A**, between a closing position and an opening position. The pack-side seal **105** closes the discharging port **102a** of the nozzle **102** when the pack-side shutter **103** is at the closing position, and the discharging port **102a** is open when the pack-side shutter **103** is at the opening position. When the pack-side shutter **103** is at the opening position, the discharging port **102a** of the nozzle **102** is exposed through the opening **103a**.

(90) FIGS. **10A** and **10B** illustrate the state where the pack-side shutter **103** is at the closing

position and the state where the pack-side shutter **103** is at the opening position, respectively. As FIG. **10A** illustrates, when the pack-side shutter **103** at the closing position is rotated about the rotational axis A in an allow K direction (first rotational direction), the pack-side shutter **103** reaches the opening position illustrated in FIG. **10B**. Conversely, when the pack-side shutter **103** is rotated in an allow L direction (second rotational direction) from the opening position, the pack-side shutter **103** reaches the closing position. In the rotation operation of the pack-side shutter **103**, the pack-side shutter **103** slide with the side surface **102c** of the nozzle **102** with the pack-side seal **105** therebetween.

(91) The detailed configurations of the nozzle **102** and the pack-side shutter **103** will be described using FIGS. **15** to **17**. An arrow N direction is a direction from the storage portion **101** toward the nozzle **102**, and an arrow U direction is a direction opposite to the arrow N direction. The arrow N direction and the arrow U direction are parallel to the rotational axis A.

(92) FIG. **15A** is an enlarged view of the vicinity of the nozzle **102** when the pack-side shutter **103** is at the closing position. FIG. **15B** illustrates the toner pack **100** viewed in the arrow U direction in FIG. **15A**. FIG. **16A** is an enlarged view of the vicinity of the nozzle **102** when the pack-side shutter **103** is at the opening position. FIG. **16B** illustrates the toner pack **100** viewed in the arrow U direction in FIG. **16A**. FIG. **17** illustrates the vicinity of the nozzle **102** viewed from the side opposite from that of FIG. **14**.

(93) As FIGS. **15A** and **15B** illustrate, the nozzle **102** includes a positioned portion **102d** having a surface **102d1** (first nozzle surface, first opposed surface) and a surface **102d2** (second nozzle surface, second opposed surface). The surface **102d1** and the surface **102d2** are spaced in an arrow R direction (the second direction D2) so as to be opposed to each other and extend in a direction crossing the R direction. As FIG. **15B** illustrates, the surface **102d1** and the surface **102d2** in the present embodiment extend in a direction perpendicular to the arrow R direction and are parallel to each other. That is, the arrow R direction is the normal direction of each of the surface **102d1** and the surface **102d2**. The positioned portion **102d** is engaged with the positioning portion **107a** (FIG. **4A**) of the first frame **107** when the toner pack **100** is mounted on the installation portion **106**. Thus, the position of the nozzle **102** relative to the first frame **107** (the base frame **2**) in the arrow R direction is determined. Accordingly, the position of the nozzle **102** relative to the first frame **107** in a rotational direction around the rotational axis A is also determined. In FIG. **15B**, a straight line CL1 (first imaginary straight line), while extending through the center in the R direction between the surface **102d1** and the surface **102d2**, extends in a direction perpendicular to the arrow R direction, and the straight line CL1 is in a phase rotated by approximately 90 degrees relative to a straight line CL2 (second imaginary straight line) passing across the rotational axis A and extending through the center of the discharging port **102a**. That is, the straight line CL1 and the straight line CL2 are orthogonal to each other.

(94) In addition, as FIGS. **15** and **18** illustrate, in the direction of the rotational axis A, a surface **102e1** and a surface **102e2** are provided on the downstream side relative to the surface **102d1** and the surface **102d2** in the N direction, respectively. As FIG. **15B** illustrates, the surface **102e1** and the surface **102e2** extend in the radial direction r of the imaginary circle VC centered around the rotational axis A.

(95) In FIG. **18**, a side surface **102e3** (second outer surface) is provided, in the arrow R direction, between the surface **102d1** and the surface **102d2** and between the surface **102e1** and the surface **102e2**. The side surface **102e3** is recessed inward in the radial direction r relative to the side surface **102c**. A recess **102e** (nozzle recess) is formed by the surface **102d1**, the surface **102d2**, and the side surface **102e3**, and the surface **102e1**, the surface **102e2**, and the side surface **102e3**.

(96) Note that the surface **102d1** and the surface **102d2** do not necessarily parallel as in the present embodiment. The surface **102d1** and the surface **102d2** may each be a surface extending in the radial direction r of the imaginary circle VC centered around the rotational axis A.

(97) In addition, as FIG. **15** illustrates, when viewed in a direction orthogonal to the direction of the

rotational axis A (the first direction D1), a side surface **103d** (first outer surface of rotary member) of the pack-side shutter **103** has the opening **103a** (rotating-body opening). In FIG. 15A, when the pack-side shutter **103** is at the closing position, at least a portion of the recess **102e** of the nozzle **102** is exposed through the opening **103a**. This is because, when the toner pack **100** in the state where the pack-side shutter **103** is closed is mounted on the installation portion **106**, the recess **102e** (the surface **102d1** and the surface **102d2**) is caused to be engaged with the positioning portion **107a**.

(98) Moreover, FIG. 15B is a bottom view of the nozzle **102** and the pack-side shutter **103** when the pack-side shutter **103** is at the closing position. When viewed in the direction of the rotational axis A (the first direction D1), a driven transmission portion **103e** (rotating-body recess) is provided across the rotational axis A from the recess **102e** (the opening **103a** of the pack-side shutter **103**) of the nozzle **102**. The driven transmission portion **103e** has a surface **103b1** and a surface **103b2** that are both extend in a direction perpendicular to the arrow R direction. FIG. 17 is an enlarged perspective view of the vicinity of the pack-side shutter **103** viewed from the side where the driven transmission portion **103e** is provided. Between the surface **103b1** and the surface **103b2**, there is provided a side surface **103b3** (second outer surface of rotary member) recessed inward in the radial direction r relative to the side surface **103d**. The driven transmission portion **103e** is constituted by the surface **103b1**, the surface **103b2**, and the side surface **103b3**.

(99) When the pack-side shutter **103** is rotated in the arrow K direction from the closing position illustrated in FIG. 15, the pack-side shutter **103** takes the opening position as FIG. 16 illustrates, and the discharging port **102a** of the nozzle **102** is exposed through the opening **103a** of the pack-side shutter **103**.

(100) As FIGS. 15 and 17 illustrate, the pack-side shutter **103** includes radial-direction positioning portions **103f** protruding outward in the radial direction r relative to the side surface **103d**. The radial-direction positioning portions **103f** are provided on the upstream side of the pack-side shutter **103** in the N direction in the direction of the rotational axis A. The radial-direction positioning portions **103f** are spaced at three spots in the rotational direction of the pack-side shutter **103** (circumferential direction of the imaginary circle VC). When the toner pack **100** is mounted on the installation portion **106**, the radial-direction positioning portions **103f** of the pack-side shutter **103** contact the inner peripheral surface **109h** of the apparatus-side shutter **109**, thereby determining the position of the toner pack **100** in the radial direction r.

(101) Note that the nozzle **102** in the present embodiment is a part including the passage **102h** through which the toner passes and the discharging port **102a** allowing the toner to be discharged from the nozzle **102**. The cross-sectional area of the passage, of the nozzle **102**, through which the toner passes may decrease or increase toward the discharging port **102a** or may be uniform. The cross-sectional area and the length of the passage of the nozzle **102** may be appropriately changed according to, for example, performance of discharging toner and are thus not limited. In addition, the discharging port **102a** of the nozzle **102** is not necessarily the most downstream opening through which the toner is discharged from the toner pack **100**. The toner discharged from the discharging port **102a** of the nozzle **102** may pass through a passage in a member different from the nozzle **102** and may then be discharged outside the toner pack **100**. In addition, although formed by polypropylene that is not flexible in the present embodiment, the nozzle **102** may be formed by another material such as ABS or polystyrene.

(102) Note that the pack-side shutter **103** may be a rotating body including the driven transmission portion **103e** and allowing the discharging port **102a** of the nozzle **102** to be always open regardless the rotation position of the pack-side shutter **103**. In this case, the discharging port **102a** of the nozzle **102** may be closed with a seal when the toner pack **100** is not yet mounted on the installation portion **106**, and the seal may be removed by a mounting operation on the installation portion **106** or may be removed after the toner pack **100** is mounted. In addition, the toner pack **100** without the pack-side shutter **103** may be possible.

(103) Supply Operation

(104) Next, an operation for mounting the toner pack **100** and supplying the toner will be described using FIGS. **19** to **25**.

(105) FIG. **19A** is a perspective view of the toner pack **100** and the installation portion **106** in the middle of the mounting of the toner pack **100** on the installation portion **106**. FIG. **19B** is a perspective view of the toner pack **100** and the installation portion **106** from a point of view different from that of FIG. **19A**. FIG. **20** is a sectional view parallel to the rotational axis A (the rotational axis B) and illustrates a mounting completion state where the toner pack **100** in the state of FIG. **19** has been moved further in the mounting direction. FIG. **21A** is a sectional view taken along line XXIA-XXIA in FIG. **20**. FIG. **21B** is a sectional view taken along line XXIB-XXIB in FIG. **20**. FIG. **22A** is a perspective view of the toner pack **100** viewed from above when the operation lever **108** is at the opening position. FIG. **22B** is a perspective view illustrating the pack-side shutter **103** and the apparatus-side shutter **109** in an opening state changed from the state of FIG. **22A** by operating the operation lever **108**. FIG. **22C** is a schematic view illustrating how a user supplies the toner. FIG. **1A** is a sectional view, taken along line IA-IA in FIG. **20**, illustrating a state changed from the state of FIG. **20** after the operation lever **108** is operated. FIGS. **1B** and **1C** are sectional views illustrating the state where the toner pack **100** in the state of FIG. **22A** is flattened when a user supplies the toner.

(106) As FIG. **19** illustrates, the toner pack **100** in the state where the pack-side shutter **103** is at the closing position is moved in the mounting direction M to be mounted on the installation portion **106** in the state where the apparatus-side shutter **109** is at the closing position. At this time, when viewed in the mounting direction M, in the rotational direction of the pack-side shutter **103**, the position of the recess **102e** (the opening **103a** of the pack-side shutter **103**) of the nozzle **102** and the position of the positioning portion **107a** of the first frame **107** are aligned. At the same time, in the rotational direction of the pack-side shutter **103**, the position of the driven transmission portion **103e** of the pack-side shutter **103** and the position of the drive transmission portion **108a** of the operation lever **108** are also aligned.

(107) After the above-described positioning is performed, when the mounting on the installation portion **106** proceeds while the toner pack **100** is moved in the mounting direction M, the toner pack **100** reaches a mounting completion position illustrated in FIG. **20**. As FIG. **20** illustrates, a small diameter portion **109d2** of the center boss **109d** of the apparatus-side shutter **109** is fitted along (engaged with) an inner peripheral surface **102b1** of a projection **102b** of the nozzle **102**. Thus, in a lower portion (the downstream side in the mounting direction M) of the nozzle **102**, the position of the nozzle **102** in the radial direction relative to the apparatus-side shutter **109** is determined. In addition, the radial-direction positioning portions **103f** (FIG. **14**, FIG. **16**) of the pack-side shutter **103** at three spots are in contact with the inner peripheral surface **109h** (FIG. **8**) of the apparatus-side shutter **109**.

(108) Thus, on the upstream side in the mounting direction M, the positions of the nozzle **102** and the pack-side shutter **103** (the toner pack **100**) in the radial direction are determined. On the other hand, a projection end surface **102b2** of the projection **102b** of the nozzle **102** is in contact with the pack contact surface **109g** of the apparatus-side shutter **109**. Thus, the position of the nozzle **102** (the toner pack **100**) in the direction of the rotational axis A (the mounting direction M) relative to the installation portion **106** is determined. Moreover, as FIG. **21A** illustrates, the positioning portion **107a** of the first frame **107** engages with the positioned portion **102d**, of the nozzle **102**, having the surface **102d1** and the surface **102d2**. Thus, in the arrow R direction of the surface **102d1** and the surface **102d2**, the position of the nozzle **102** is determined relative to the first frame **107** (the base frame 2). Note that the inner peripheral surface **102b1** of the projection **102b** is not necessarily fitted onto the center boss **109d** as long as there is no interference. As FIG. **21A** illustrates, the drive transmission portion **108a** (protruded portion of lever) of the operation lever **108** engages with the driven transmission portion **103e** (rotating-body recess, refer to FIG. **17**) of

the pack-side shutter **103**. Moreover, as FIG. **21B** illustrates, the driven transmission portion **103e** (rotating-body recess) of the pack-side shutter **103** engages with the driven transmission portion **109e** (protruded portion of shutter) of the apparatus-side shutter **109**. Thus, the rotational axis A of the pack-side shutter **103** and the rotational axis B of the apparatus-side shutter **109** are substantially coaxial with each other. Relative to the first frame **107** (the base frame **2**) and the nozzle **102**, the operation lever **108**, the pack-side shutter **103**, and the apparatus-side shutter **109** are movable as a substantially integrated body in the rotation about the rotational axis A (the rotational axis B). Specifically, when the operation lever **108** is rotated, the drive transmission portion **108a** of the operation lever **108** pushes the surface **103b1** or **103b2** of the pack-side shutter **103** to rotate the pack-side shutter **103**. Subsequently, the surface **103b1** or the surface **103b2** of the pack-side shutter **103** pushes the driven transmission portion **109e** of the apparatus-side shutter **109** to rotate the apparatus-side shutter **109**.

(109) As FIGS. **22A** and **22B** illustrate, after the mounting of the toner pack **100** on the installation portion **106** is completed, the operation portion **108b** of the operation lever **108** is rotated in a rotational direction D. When the operation lever **108** is operated so as to rotate in the rotational direction D, the apparatus-side shutter **109** rotates from the closing position to the opening position, and the pack-side shutter **103** rotates from the closing position to the opening position.

(110) Action of Loosening Member Accompanying Supply Operation

(111) The action of the loosening member **200** accompanying the supply operation with the toner pack **100** will be described.

(112) As FIG. **22C** illustrates, as the supply operation, a user presses a region H that is a part of the storage portion **101** of the toner pack **100**. The region H is, as FIG. **10** illustrates, a region in which the first loosening portion **202**, which is disposed inside the storage portion **101**, is disposed. As described above, the loosening member **200** is supported by the nozzle **102** so as to be swingable relative to the nozzle **102** about the supported portion **201b1** (the swing axis S1). Moreover, the second loosening portion **201b2** is also swingable relative to the nozzle **102** about the supported portion **201b1**. Thus, as FIG. **1A** illustrates, the first loosening portion **202** is elastically deformed in an arrow S direction by a pressing force F1 and a pressing force F2 that press the storage portion **101** from both outer sides in the second direction D2, and the entire loosening member **200** also swings in an arrow T direction or an arrow V direction about the supported portion **201b1**. When the pressing force F1 is greater than the pressing force F2, the toner pack **100** transitions from the state of FIG. **1A** to the state of FIG. **1B**. When the pressing force F2 is greater than the pressing force F1, the toner pack **100** transitions from the state of FIG. **1A** to the state of FIG. **1C**. In addition, when the user stops applying the pressing force F1 and the pressing force F2, the toner pack **100** returns to the state of FIG. **1A** or a state close thereto with the restoration of the first loosening portion **202** from the state of being elastically deformed. Due to such an above-described action of the loosening member **200**, the toner stored in the storage portion **101** is loosened by the first loosening portion **202**. At the same time, the toner inside the passage **102h** of the nozzle **102** is loosened by the second loosening portion **201b2**. Due to the own weight of the toner and increase in the internal pressure of the storage portion **101** caused by pressing the storage portion **101**, the toner inside the storage portion **101** and the toner inside the passage **102h** are discharged through the discharging port **102a** of the nozzle **102** along arrow Z. When the supply operation ends, the operation lever **108** is rotated in a rotational direction E by operating the operation portion **108b**, and the toner pack **100** is then removed from the installation portion **106**.

(113) With the toner pack **100** according to the present embodiment, the toner inside the storage portion **101** and the toner inside the passage **102h** can be loosened efficiently at the same time of the supply operation, without loosening the toner before the supply, compared with when no loosening member **200** is provided. Moreover, the toner pack **100** is improved in performance of discharging toner, and the toner remaining in the toner pack **100** after the supply can thereby be reduced. In addition, due to the elastic restorative force of the first loosening portion **202**, in the

storage portion **101**, a certain volumetric size or more is maintained, with which the storage portion **101** is easily flattened. For example, even when the toner pack **100** is flattened by, for example, impact during physical distribution and temporarily brought into a state where the air therein is discharged through the pack-side seal **105**, the shape of the toner pack **100** is easily restored.

(114) Note that the storage portion **101** of the toner pack **100** is not limited to a pouch as in the present embodiment and may be a container made of, for example, paper or vinyl. In addition, as FIG. 23A illustrates, an elastic portion **1010a** that partially elastically deforms with ease may be provided in an end portion, on the further side from the nozzle **102** in the first direction **D1**, of a storage portion **1010** (bottle) made of resin and having a thickness of 0.5 mm or more and 2.0 mm or less. The elastic portion **1010a** is, for example, a thin portion having a thickness (thickness of 0.1 mm or more and 0.3 mm or less) smaller than the part other than the elastic portion **1010a**, or a portion constituted by a material having a Young's modulus smaller than the material of the part other than the elastic portion **1010a**. The elastic portion **1010a** is elastically deformed by being pressed by a user as FIG. 23B illustrates, and the pressing force **F1** applied by the user is transmitted to the loosening member **200** inside the storage portion **1010**; thus, the toner can be loosened. In addition, as in FIG. 24A, an elastic portion **1020b** that partially elastically deforms with ease may be formed in an end portion, on the nozzle **102** side in the first direction **D1**, of a storage portion **1020** (bottle) made of resin and having a thickness of 0.5 mm or more and 2.0 mm or less. The elastic portion **1020b** is, for example, a thin portion having a thickness (thickness of 0.1 mm or more and 0.3 mm or less) smaller than the part other than the elastic portion **1020b**, or a portion constituted by a material having a Young's modulus smaller than the material of the part other than the elastic portion **1020b**. The storage portion **1020** is moved relative to the nozzle **102** by a pressing force **F3** applied by the user, and the loosening member **200** receives a pressing force **F4** from the storage portion **1020** and can thus loosen the toner.

(115) In addition, although the loosening member **200** is supported so as to be swingable relative to the nozzle **102**, and the first loosening portion **202** is an elastically deformable wire spring and has a helical shape in the present embodiment, this is not the only option. For example, the first loosening portion **202** and the second loosening portion **201b2** may be a loosening member **2000** in FIG. 25A having a comb shape or may be a loosening member **2000** in FIG. 25B having a mesh shape. The loosening member **2000** in FIG. 25A includes a first loosening portion **2020** and a second loosening portion **2030**. The first loosening portion **2020** is constituted by plural protrusions extending in the second direction **D2** and aligned in the first direction **D1**. The second loosening portion **2030** is constituted by plural protrusions shorter than the protrusions of the first loosening portion **2020** and aligned in the first direction **D1**. The loosening member **2000** in FIG. 25B includes a first loosening portion **2020** and a second loosening portion **2030** each having a mesh shape.

(116) In addition, the loosening member **200** is not necessarily supported by the nozzle **102** and may be kept floating inside the storage portion **101**. The base portion **201** itself may be formed by a wire spring. When the toner in the passage **102h** of the nozzle **102** is not required to be loosened, the second loosening portion **201b2** is not necessarily provided. With the configuration in which the pressing force applied by a user is transmitted to the loosening member **200**, the toner stored in the storage portion **101** can be loosened by moving the position of the loosening member **200** relative to the storage portion **101** and the nozzle **102**.

Second Embodiment

(117) The configuration of a second embodiment will be described. Note that the description of the same elements as those in the above-described embodiment may be omitted. In addition, the elements corresponding to the elements described in the first embodiment may be denoted by names similar to those of the elements in the first embodiment. In the present embodiment, the elements different from those in the first embodiment will mainly be described.

(118) The configuration of a toner pack **250** in the present embodiment will be described using

FIGS. 26 to 29. Note that the configuration of the installation portion 106 is the same as that in the first embodiment. FIG. 26A is a front view of the toner pack 250. FIG. 26B is a perspective view of the toner pack 250. FIG. 27A is an enlarged view of the vicinity of a nozzle 252 when a pack-side shutter 253 is at a closing position.

(119) FIG. 27B is an enlarged view of the vicinity of the nozzle 252 when the pack-side shutter 253 is at an opening position. FIG. 28A is a perspective view of a loosening member 254. FIG. 28B is a side view of the loosening member 254. FIG. 28C is a top view of the loosening member 254. FIG. 29 is a sectional view of the toner pack 250 in FIG. 26A, taken along line XXIX-XXIX. FIG. 30A is a sectional view parallel to a rotational axis A (the rotational axis B) in the state where the mounting is completed. FIG. 30B is a section taken along line XXXB-XXXB in FIG. 30A. FIG. 31 is a sectional view parallel to the rotational axis A (the rotational axis B) with the pack-side shutter 253 and the apparatus-side shutter 109 being rotated to be brought into an opening state. Note that the loosening member 254 is not illustrated in a form of section in FIGS. 29, 30A, and 31 for good viewability.

(120) The toner pack 250 includes a storage portion 251 storing the toner, the nozzle 252, the pack-side shutter 253, and the loosening member 254. As FIG. 26 illustrates, in the first direction D1, the storage portion 251 is provided on one end side, and the nozzle 252 and the pack-side shutter 253 are provided on the other end side. The storage portion 251 is a bottle made of resin. The resin bottle in the present embodiment has a thickness of 0.5 mm or more and 2.0 mm or less. Examples of the material include polystyrene (PS), polyethylene (PE), polyethylene terephthalate (PET), PC, ABS, PC+ABS, PP, and nylon. As FIG. 27 illustrates, the pack-side shutter 253 is rotatable about the rotational axis A, between the closing position and the opening position. The pack-side seal 105 closes a discharging port 252c of the nozzle 252 when the pack-side shutter 253 is at the closing position, and the discharging port 252c is open when the pack-side shutter 253 is at the opening position. A projection 253a is provided at a leading end of the pack-side shutter 253.

(121) As FIGS. 28A and 28B illustrate, the loosening member 254 includes a base portion 255, a first loosening portion 256, and a second loosening portion 257. The first loosening portion 256 is constituted by comb-shaped parts provided on the base portion 255, the parts are arranged every 90 degrees in the rotational direction around the rotational axis A, and the parts are provided in a staggered manner in the direction of the rotational axis A. That is, the first loosening portion 256 has first protruding portions extending in the second direction D2 and second protruding portions extending in the third direction crossing the first direction D1 and the second direction D2. The second loosening portion 257 is a comb-shaped part provided on the base portion 255.

(122) As FIG. 29 illustrates, the loosening member 254 is provided inside the storage portion 251. The second loosening portion 257 is disposed inside a passage 252a of the nozzle 252. A leading end 255a of the base portion 255 is fitted in and fixed to a hole portion 253b of the pack-side shutter 253.

(123) Next, a mechanism by which the loosening member 254 loosens the toner will be described using FIGS. 30 to 32. FIG. 30A is a sectional view illustrating the state where the toner pack 250 is mounted on the installation portion 106. As in the above-described embodiment, when, in the state of FIG. 30A, the operation lever 108 is rotated in the rotational direction D by operating the operation portion 108b (refer to FIGS. 22A and 22B), each of the apparatus-side shutter 109 and the pack-side shutter 253 is rotated from the closing position to the opening position. Because the loosening member 254 is fixed to the pack-side shutter 253, the loosening member 254 rotates about the rotational axis A in the arrow K direction. Thus, the toner in the storage portion 251 is loosened by the first loosening portion 256, and the toner in the passage 252a is loosened by the second loosening portion 257. As a result, as FIG. 31 illustrates, the loosened toner is discharged from the storage portion 251 along arrow Z due to the own weight. In other words, the toner in the storage portion 251 is loosened by a shutter opening and closing operation, and the toner is thereby easily discharged. After the toner supply ends, the operation portion 108b is rotated in a rotational

direction E, and the toner pack **250** is then removed from the installation portion **106**.

(124) Note that, although the storage portion **251** of the toner pack **250** is a resin bottle in the present embodiment, this is not the only option. For example, a pouch formed by a sheet such as a flexible polypropylene sheet being subjected to pouch processing may be possible. In addition, the form of the loosening member **254** is not necessarily limited thereto. For example, as FIG. **32** illustrates, a first loosening portion **256** may be a screw including a blade portion having a helical shape, and a second loosening portion **257** may be an elastically deformable sheet.

Third Embodiment

(125) Next, a third embodiment will be given. Note that the description of the elements similar to those in the above-described embodiments may be omitted. Of the elements in the present embodiment, the elements corresponding to those in the first embodiment may be denoted by names similar to those of the elements in the first embodiment. In the present embodiment, the elements different from those in the first embodiment will be described.

(126) First, the configuration of an installation portion **300** in the present embodiment will be described using FIGS. **33** to **34**. FIG. **33** is a perspective view of the installation portion **300**. FIG. **34A** is a top view of the installation portion. FIG. **34B** is a section taken along line XXXIVB-XXXIVB in FIG. **34A**.

(127) As FIG. **33** and FIG. **34A** illustrate, a drive transmission member **302** is provided on a leading end portion of a center boss **301a** of an apparatus-side shutter **301** constituting the installation portion **300**. The drive transmission member **302** is provided with four drive transmission claws **302a**. As FIG. **34B** illustrates, a drive shaft **303** is fixed to the drive transmission member **302** and is rotatable about a rotational axis B in the rotational direction D by receiving a driving force from a drive source M (motor).

(128) Next, the configuration of a toner pack **350** in the present embodiment will be described using FIGS. **35** to **37**. FIG. **35A** is a front view of the toner pack **350**. FIG. **35B** is a perspective view of the toner pack **350**. FIG. **36A** is a front view of a loosening member **354**. FIG. **36B** is a perspective view of the loosening member **354**. FIG. **37A** is a section taken along line XXXVIIA-XXXVIIA in FIG. **35A**. The arrow U direction is a direction from a nozzle **352** toward a storage portion **351**. FIG. **37B** illustrates the configuration in FIG. **37A** viewed in the arrow U direction.

(129) As in the first embodiment, the toner pack **350** includes the storage portion **351** storing the toner, the nozzle **352**, a pack-side shutter **353**, and the loosening member **354**. As FIG. **35** illustrates, in the first direction D1, the storage portion **351** is provided on one end side, and the nozzle **352** and the pack-side shutter **353** are provided on the other end side. The storage portion **351** is formed by a resin bottle similar to that in the second embodiment.

(130) As FIG. **36** illustrates, the loosening member **354** includes a shaft portion **355** rotatable about a rotational axis A, a first loosening portion **356** provided on an outer peripheral surface of the shaft portion **355**, and a second loosening portion **357** having an outside diameter smaller than the first loosening portion **356** and provided on the outer peripheral surface of the shaft portion **355**. The first loosening portion **356** and the second loosening portion **357** are each a blade (blade portion) having a helical shape whose helical axis coincides with the rotational axis. That is, the loosening member **354** is a screw. As FIG. **37A** illustrates, the loosening member **354** is disposed inside the storage portion **351**. A driven transmission member **358** (driving-force receiving portion) is fixed to a first end portion **355a** provided at one end of the shaft portion **355**. The driven transmission member **358** is a coupling provided with four driven transmission claws **358a** that are provided on an end surface in the direction of the rotational axis A.

(131) When, with the nozzle **352** being positioned below the storage portion **351**, the toner pack **350** is oriented in a predetermined direction such that the rotational axis A is parallel to the gravitational direction, the driven transmission member **358** is fixed to the first end portion **355a** (lower end portion) that is an end portion of the shaft portion **355** on the lower side. As FIGS. **37A** and **37B** illustrate, the driven transmission member **358** is exposed outside the toner pack **350** from

a bottom surface of the nozzle **352** when the toner pack **350** is oriented in the predetermined direction.

(132) The shaft portion **355** includes a portion (first portion) positioned inside the storage portion **351** and provided with the first loosening portion **356** and a portion (second portion) positioned inside a passage **352a** and provided with the second loosening portion **357**. The first end portion **355a** that is one end portion of the shaft portion **355** is, as FIG. **38** illustrates, a leading end portion penetrating an inner wall of the passage **352a** downward.

(133) A second end portion **355b** that is the other end portion of the shaft portion **355** and a received portion **355c** are supported by a bearing **351a** of the storage portion **351** and a bearing **352b** of the nozzle **352**, respectively, so as to be rotatable about the rotational axis A. The bearing **351a** is provided on a top surface inside the storage portion **351** when the toner pack **350** is oriented in the predetermined direction. The second loosening portion **357** is disposed inside the passage **352a** of the nozzle **352**.

(134) Next, a supply operation of toner will be described using FIGS. **38** to **40**. FIG. **38** is a sectional view illustrating the state where the toner pack **350** is mounted on the installation portion **300**, the operation lever **108** is rotated in the rotational direction D (refer to FIGS. **22A** and **22B**), and each of the apparatus-side shutter **301** and the pack-side shutter **353** is rotated from a closing position to an opening position. FIG. **39** is a perspective view illustrating how the drive transmission member **302** attached to the drive shaft **303** and the driven transmission member **358** attached to the loosening member **354** mesh with each other.

(135) As FIGS. **38** and **39** illustrate, with the toner pack **350** being mounted on the installation portion **300**, the drive transmission member **302** attached to the drive shaft **303** and the driven transmission member **358** attached to the loosening member **354** mesh with each other (engage with each other). Due to the contact between the drive transmission claws **302a** of the drive transmission member **302** and the driven transmission claws **358a** of the driven transmission member **358**, when the drive shaft **303** is rotated in the rotational direction D by the drive source M, the loosening member **354** is also rotated in the rotational direction D. Thus, the toner in the storage portion **351** of the toner pack **350** is loosened by the first loosening portion **356**, and the toner in the passage **352a** of the nozzle **352** is loosened by the second loosening portion **357**. The toner in the storage portion **351** is then discharged through a discharging port **352c** along arrow Z due to the own weight of the toner and the conveyance force of the loosening member **354**.

(136) In other words, the loosening member **354** is rotated by receiving a driving force from the image forming apparatus **1**, thereby loosening the toner inside the storage portion **351** and the toner inside the passage **352a** of the nozzle **352**. After the toner supply ends, the operation lever **108** is rotated in the rotational direction E by operating the operation portion **108b**, and the toner pack **350** is then removed from the installation portion **300**.

(137) Note that, although the first loosening portion **356** and the second loosening portion **357** of the loosening member **354** are each a screw in the present embodiment, this is not the only option. For example, as FIG. **40** illustrates, a flexible sheet of, for example, PET or PC may be possible. The free lengths of the first loosening portion **356** and the second loosening portion **357** are made larger than the inside diameter of the storage portion **351**, and the first loosening portion **356** and the second loosening portion **357** thereby slide with the inner wall of the storage portion **351**; thus, the toner in the entire storage portion **351** can be loosened efficiently. In addition, although the storage portion **351** is a resin bottle in the present embodiment, this is not the only option. A pouch formed by a sheet such as a flexible polypropylene sheet being subjected to pouch processing may be possible. In addition, the loosening member **354** is not necessarily a screw as long as the loosening member **354** can loosen the toner in the storage portion **351** by moving.

(138) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all

such modifications and equivalent structures and functions.

(139) This application claims the benefit of Japanese Patent Application No. 2023-106710, filed Jun. 29, 2023, which is hereby incorporated by reference herein in its entirety.

Claims

1. A toner container comprising: a storage portion configured to store toner; a nozzle portion having a discharging port configured to discharge, outside the toner container, toner stored in the storage portion, the nozzle portion being fixed to the storage portion such that the nozzle portion and the storage portion are aligned in a first direction; and a loosening member configured to loose toner, the loosening member including at least a portion provided in the storage portion, wherein the loosening member is swingable relative to the storage portion about a swing axis extending in a second direction crossing the first direction.
2. The toner container according to claim 1, wherein the loosening member includes: a base portion whose longitudinal direction coincides with the first direction, the base portion including a first end portion that is one end portion in the longitudinal direction and a second end portion that is the other end portion in the longitudinal direction; and an elastic portion elastically displaceable relative to the base portion, the elastic portion being supported by the base portion so as to be positioned on an outer side relative to the base portion in a direction crossing the first direction.
3. The toner container according to claim 2, wherein the second end portion is provided with a supported portion supported by the nozzle portion such that the loosening member is swingable relative to the nozzle portion about the second direction.
4. The toner container according to claim 3, wherein the nozzle portion includes a passage allowing toner to pass through toward the discharging port, and wherein a leading end portion provided in the second end portion and provided at a position further apart from the first end portion than the supported portion is positioned in the passage.
5. The toner container according to claim 4, wherein the discharging port is provided in a side surface, of the nozzle portion, extending in the first direction, and wherein the leading end portion extends in a direction where a distance from the discharging port in a direction crossing the first direction decreases as a distance from the discharging port in the first direction decreases.
6. The toner container according to claim 4, wherein the leading end portion is provided with a plurality of protrusions protruding in a direction crossing a direction where the leading end portion extends toward the discharging port.
7. The toner container according to claim 2, wherein the first end portion of the base portion is a free end in the storage portion.
8. The toner container according to claim 2, wherein the elastic portion is a wire material extending in the first direction while turning around the base portion, and wherein a third end portion and a fourth end portion that are one end portion and the other end portion of the wire material, respectively, are fixed to the first end portion and the second end portion of the base portion, respectively.
9. The toner container according to claim 8, wherein the storage portion includes a region whose width in a direction crossing the first direction increases as a distance from the nozzle portion in the first direction increases, and wherein the wire material includes, in the region, a portion whose radius of turning around the base portion increases as the distance from the nozzle portion in the first direction increases.
10. The toner container according to claim 2, wherein the first end portion of the base portion is a free end extending in a direction crossing the first direction.
11. The toner container according to claim 2, wherein the elastic portion includes a plurality of protruding portions protruding from the base portion in a direction crossing the first direction and aligned in the first direction.

12. The toner container according to claim 2, wherein the elastic portion includes a first protruding portion protruding from the base portion in a direction crossing the first direction and a second protruding portion protruding in a direction crossing the first direction and the direction where the first protruding portion protrudes.
13. The toner container according to claim 2, wherein the storage portion includes a region whose width in a third direction crossing the first direction increases as a distance from the nozzle portion in the first direction increases, and wherein the elastic portion includes, in the region, a portion whose width in the third direction increases as the distance from the nozzle portion in the first direction increases.
14. The toner container according to claim 1, wherein the nozzle portion includes a passage allowing toner to pass through toward the discharging port, and wherein an end portion of the loosening member closer to the nozzle portion is positioned in the passage of the nozzle portion.
15. The toner container according to claim 1, wherein the loosening member does not swing about an imaginary line extending in a third direction crossing the first direction and the second direction.
16. A toner container comprising: a storage portion configured to store toner; a nozzle portion having a discharging port configured to discharge, outside the toner container, toner stored in the storage portion, the nozzle portion being fixed to the storage portion such that the nozzle portion and the storage portion are aligned in a first direction; and a stirring member configured to stir toner in the storage portion, wherein the stirring member includes: a shaft portion supported by the nozzle portion so as to be rotatable about a rotational axis relative to the nozzle portion, the shaft portion being provided such that the rotational axis is oriented in the first direction; a blade portion provided on the shaft portion so as to rotate with the shaft portion; and a driving-force receiving portion configured to receive, from outside the toner container, a driving force for rotating the driving-force receiving portion, and wherein, when, with the nozzle portion being positioned below the storage portion, the toner container is oriented in a predetermined direction such that the rotational axis is parallel to a gravitational direction, the driving-force receiving portion is provided on a lower end portion that is an end portion of the shaft portion on a lower side.
17. The toner container according to claim 16, wherein the driving-force receiving portion is a coupling.
18. The toner container according to claim 17, wherein, when the toner container is oriented in the predetermined direction, the driving-force receiving portion is exposed outside the toner container from a bottom surface of the nozzle portion.
19. The toner container according to claim 17, wherein, when the toner container is oriented in the predetermined direction, the discharging port is provided in a side surface, of the nozzle portion, extending along the rotational axis, and the nozzle portion has a receiving inlet facing upward so as to receive toner from the storage portion and provided above the discharging port and includes a passage allowing toner to pass through from the receiving inlet to the discharging port, wherein the shaft portion includes a first portion positioned in the storage portion and a second portion positioned in the passage, and wherein, when the toner container is oriented in the predetermined direction, the lower end portion is a portion, of the shaft portion, penetrating the passage downward.
20. The toner container according to claim 19, wherein the blade portion is provided on each of the first portion and the second portion of the shaft portion, and wherein the blade portion of the second portion has a width smaller than the blade portion of the first portion in a direction orthogonal to the rotational axis.
21. The toner container according to claim 19, wherein the blade portion has a helical shape whose helical axis coincides with the rotational axis of the shaft portion, and the blade portion is provided on each of the first portion and the second portion of the shaft portion, and wherein the blade portion of the second portion has an outside diameter smaller than the blade portion of the first portion.

22. The toner container according to claim 16, wherein the driving-force receiving portion is positioned below the discharging port.

23. The toner container according to claim 16, wherein the blade portion has a helical shape whose helical axis coincides with the rotational axis of the shaft portion.

24. A toner container comprising: a storage portion configured to store toner; a nozzle portion having a discharging port configured to discharge, outside the toner container, toner stored in the storage portion, the nozzle portion being fixed to the storage portion such that the nozzle portion and the storage portion are aligned in a first direction; a shutter configured to rotate about a rotational axis extending in the first direction, between a closing position where the shutter closes the discharging port and an opening position where the shutter opens the discharging port; and a loosening member including at least a portion provided in the storage portion, the loosening member being configured to loose toner, the loosening member including a shaft portion centered around the rotational axis and a loosening portion provided on an outer peripheral surface of the shaft portion, the loosening member being configured to be rotated by rotation of the shutter.

25. A toner container comprising: a storage portion configured to store toner; a nozzle portion having a discharging port configured to discharge, outside the toner container, toner stored in the storage portion, the nozzle portion being fixed to the storage portion such that the nozzle portion and the storage portion are aligned in a first direction; and a loosening member configured to loose toner, the loosening member including at least a portion provided in the storage portion, wherein the loosening member includes: a base portion whose longitudinal direction coincides with the first direction, the base portion including a first end portion that is one end portion in the longitudinal direction and a second end portion that is the other end portion in the longitudinal direction; and an elastic portion elastically displaceable relative to the base portion, the elastic portion being supported by the base portion so as to be positioned on an outer side relative to the base portion in a direction crossing the first direction.

26. The toner container according to claim 25, wherein the elastic portion is a wire material extending in the first direction while turning around the base portion, and wherein a third end portion and a fourth end portion that are one end portion and the other end portion of the wire material, respectively, are fixed to the first end portion and the second end portion of the base portion, respectively.
